

# HEALTHCARE READMISSION PREDICTION PROJECT

## 1. PROJECT OVERVIEW

The objective of this project was to analyze healthcare patient data and predict whether a diabetic patient would be readmitted to the hospital within 30 days after discharge.

This project covers the complete data analytics lifecycle, including data cleaning, exploratory data analysis, feature engineering, machine learning model development, evaluation, and dashboard creation.

Dataset Size: 101,766 Records

Tools Used:

- Python
  - Pandas
  - NumPy
  - Matplotlib
  - Seaborn
  - Scikit-Learn
  - SQL
  - Power BI / HTML Dashboard
  - GitHub
- 

## 1. BUSINESS PROBLEM

Hospital readmissions increase healthcare costs and indicate potential gaps in patient care.

The goal of this project was to identify factors influencing patient readmission and build a predictive model to help healthcare organizations identify high-risk patients.

---

## 1. DATA UNDERSTANDING

The dataset contains information about diabetic patients, including:

- Patient Demographics
  - Age
  - Gender
  - Race
- Hospital Stay Duration
- Number of Medications
- Emergency Visits
- Diagnoses
- Readmission Status

Target Variable:

Readmitted

0 = Not Readmitted Within 30 Days

1 = Readmitted Within 30 Days

---

## 1. DATA CLEANING

The following steps were performed:

1. Loaded dataset into Python using Pandas.
  2. Checked dataset dimensions and data types.
  3. Identified missing values.
  4. Replaced missing values where required.
  5. Removed duplicate records.
  6. Converted target variable into binary format.
  7. Encoded categorical variables.
  8. Removed high-cardinality diagnosis columns to reduce dimensionality.
- 

## 1. EXPLORATORY DATA ANALYSIS (EDA)

Question 1: How Many Patients Were Readmitted?

Observation:

Most patients were not readmitted within 30 days.

Question 2: Age vs Readmission

Observation:

Patients in the 70–80 age group showed the highest readmission counts and were more likely to return within 30 days.

Question 3: Gender vs Readmission

Observation:

Female patients showed slightly higher readmission counts, but there was no significant difference between male and female patients.

Question 4: Race vs Readmission

Observation:

Caucasian patients accounted for the largest number of readmissions due to their larger representation in the dataset.

Question 5: Time in Hospital

Observation:

Hospital stay duration showed little difference between readmitted and non-readmitted patients.

Question 6: Number of Medications

Observation:

Medication count did not significantly influence readmission behavior.

Question 7: Emergency Visits

Observation:

Patients with higher emergency visit counts were more likely to be readmitted.

---

#### 1. KEY INSIGHTS FROM EDA

2. Most patients were not readmitted within 30 days.
3. Age group 70–80 had the highest readmission frequency.
4. Gender showed minimal impact on readmission.
5. Caucasian patients had the highest number of readmissions.
6. Hospital stay duration did not significantly affect readmission.
7. Medication count was not strongly associated with readmission.
8. Emergency visits were strongly linked to readmission.

---

#### 1. FEATURE ENGINEERING

Several transformations were applied:

- Target variable conversion
- Categorical variable encoding
- Removal of diagnosis columns with excessive unique values
- Creation of machine-learning-ready dataset

The feature count was reduced significantly to improve model performance.

---

## 1. MACHINE LEARNING MODELS

### Model 1: Logistic Regression

Results:

Accuracy: 88.77% Recall: 0%

Observation:

The model predicted almost all patients as non-readmitted due to class imbalance.

---

### Model 2: Balanced Logistic Regression

Results:

Accuracy: 58.28% Precision: 11.5% Recall: 40.7% F1 Score: 17.9%

Observation:

The model successfully identified a larger proportion of readmitted patients and performed better for healthcare prediction.

---

### Model 3: Random Forest

Results:

Accuracy: 89% Recall: 1%

Observation:

The model heavily favored the majority class and failed to identify readmitted patients effectively.

---

## 1. MODEL SELECTION

Selected Model:

Balanced Logistic Regression

Reason:

Although overall accuracy was lower, recall improved significantly.

In healthcare prediction, recall is more important because identifying high-risk patients is critical.

---

## 1. CONFUSION MATRIX

True Negatives: 10931

False Positives: 7138

False Negatives: 1354

True Positives: 931

---

## 1. DASHBOARD DEVELOPMENT

A healthcare dashboard was created to visualize:

- Total Patients
- Readmission Rate
- Age vs Readmission
- Gender vs Readmission
- Race vs Readmission
- Emergency Visits vs Readmission
- Hospital Stay Distribution

The dashboard enables stakeholders to monitor readmission trends and identify high-risk patient groups.

---

## 1. PROJECT CONCLUSION

This project successfully analyzed over 100,000 healthcare records and identified key factors associated with patient readmission.

Key Findings:

- Older patients were more likely to be readmitted.
- Emergency visits were strong indicators of future readmission.
- Dataset imbalance affected model performance.
- Balanced Logistic Regression provided the best trade-off between identifying readmitted patients and overall model performance.

The project demonstrates practical skills in data cleaning, exploratory data analysis, machine learning, dashboarding, and business insight generation.

---

END OF REPORT